

MODELS OF GERIATRIC CARE, QUALITY IMPROVEMENT,
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Annual Wellness Visits and Timing of Advance Care Planning Among Medicare Beneficiaries With Cognitive Impairment

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ABSTRACT

Background: Advance care planning (ACP) allows older adults to document preferences for future medical care, but uptake remains limited and unequal, especially among those with mild cognitive impairment (MCI) or Alzheimer's disease and related dementias (ADRD). Since 2016, ACP discussions can be billed during the Medicare annual wellness visit (AWV), creating a policy-supported opportunity for timely ACP engagement.

Participants and Setting: We studied 959,405 Medicare beneficiaries aged ≥ 68 years newly diagnosed with MCI or ADRD in 2018, using national claims data.

Methods: We conducted a retrospective cohort study with 1:1 propensity score matching to compare beneficiaries with and without AWV receipt in 2018. Cox proportional hazards models with death as a competing risk estimated associations between AWV and ACP initiation over 4 years. Sensitivity analyses included censoring rules and time-dependent modeling. Subgroup analyses assessed variation by cognitive stage, race/ethnicity, sex, region, and education.

Results: Among 118,174 beneficiaries who received an AWV, 27.62% initiated ACP within 4 years compared with 17.46% of non-recipients (HR, 1.86; 95% CI, 1.84–1.89). Most ACP uptake among AWV recipients occurred on the same day as the visit (9.00%), with modest increases thereafter. Sensitivity analyses confirmed stronger associations when accounting for continued AWV use (HR, 2.51; 95% CI, 2.46–2.56) and with time-dependent modeling (HR, 2.27; 95% CI, 2.27–2.31). The positive association between AWV and ACP was consistent across subgroups and particularly strong among Hispanic beneficiaries and those in nonmetropolitan areas.

Conclusions: AWV receipt was associated with significantly earlier and greater ACP initiation among older adults with MCI or ADRD. Incorporating ACP into AWV workflows could enhance timely engagement, reduce disparities, and strengthen alignment of care with patient preferences. System-level interventions and clinician training may further increase same-day ACP uptake.

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Summary

- Key points
 - Receipt of an annual wellness visit (AWV) was associated with significantly earlier and greater initiation of advance care planning (ACP) among Medicare beneficiaries with mild cognitive impairment (MCI) or Alzheimer's disease and related dementias (ADRD).
 - Most ACP uptake among AWV recipients occurred on the same day as the AWV, though overall completion remained modest.
 - The positive association between AWV and ACP was consistent across subgroups, with particularly strong effects among Hispanic beneficiaries and those in nonmetropolitan areas.
- Why does this paper matter?
 - This study provides the first national evidence that AWVs can serve as a timely and equitable entry point for ACP among cognitively impaired older adults. The findings highlight opportunities to integrate ACP into AWV workflows to improve early planning and reduce disparities in end-of-life care.

1 | Introduction

Advance care planning (ACP) refers to the process by which individuals articulate and document their future medical care preferences, often before serious illness or loss of decision-making capacity [1, 2]. ACP has been linked to increased completion of advance directives, improved family communication, and greater alignment between end-of-life care and patient values [3, 4].

Timely documentation of future care preferences is especially important for individuals with mild cognitive impairment (MCI) or early Alzheimer's disease and related dementias (ADRD), an early stage of neurocognitive conditions when individuals still have decisional capacity to make informed decisions about their future care. Individuals with MCI often retain decisional capacity for many years; however, they also face an increased risk of future declines in understanding, appreciation, and reasoning abilities [5, 6]. This risk makes the early stage of impairment an important period to initiate ACP while patients can still reliably express their preferences [7]. Timely ACP within this period has been shown to reduce unnecessary hospitalization and increase the use of home- and hospice-based services [8]. However, real-world uptake of ACP remains inconsistent and unequally distributed, shaped by a complex interplay of patient-level factors such as cognitive burden, minority status, and geographic isolation; varying degrees of family engagement; provider characteristics including training, age, and concerns about reimbursement; and system-level conditions such as the presence of EMR alerts or integrated palliative care, while structural and cultural biases further contribute to persistent disparities across race, region, and socioeconomic status [9–13].

Since 2016, providers have been able to bill for ACP discussions during AWVs, offering a policy-supported opportunity for early engagement [14]. AWVs have also been linked to earlier diagnoses of MCI, further reinforcing their potential to serve as timely touchpoints for ACP [15]. However, real-world uptake of ACP during AWVs remains limited and uneven, particularly among cognitively impaired older adults [16, 17]. In the general population, national data show that only 10.7% of ACP occurred on the same day as an AWV in 2017, with only a modest increase to 14.1% by 2019 [14, 18]. Even among patients with multiple AWVs, fewer than half completed ACP during their first visit, and only 22% of the remainder did so later [16]. Among older adults with dementia, fewer than 7.5% have any billed ACP within a 12-month perioperative window, whereas when ACP is defined more broadly to include nonbilled activities such as living wills and durable powers of attorney, approximately 43%–47% of those with cognitive impairment or dementia report having completed these documents, suggesting considerable unrealized potential for ACP that AWVs could help activate [12, 19, 20]. These patterns likely reflect barriers in workflow, billing, and clinical engagement, yet prior research has not examined whether the timing of AWV affects when ACP occurs, or whether this varies across subgroups [11].

We used Medicare claims to examine the link between AWV completion and ACP timing among beneficiaries newly diagnosed with MCI or ADRD. Our study had three aims: first, to assess whether AWV receipt after diagnosis is associated with earlier ACP over 4 years; second, to evaluate whether ACP occurs on the same day as the AWV or later; and third, to test whether associations vary by race, gender, region, or education. These patterns may reveal opportunities to promote more timely and equitable ACP in cognitively impaired populations.

2 | Methods

2.1 | Source of Data

We used Medicare data from 100% beneficiaries with diagnosis of ADRD/MCI in 2018 through access to the Centers for Medicare and Medicaid Service (CMS) Virtual Research Data Center. We used the Master Beneficiary Summary File to determine beneficiary demographic characteristics. Outpatient AWVs were determined using the Outpatient Standard Analytical files and Carrier files. ACP was determined using the Medicare Provider Analysis and Review files, Outpatient Standard Analytical files, and Carrier files. The institutional review board of the University of Texas Medical Branch approved this study.

2.2 | Cohort Selection

The study population consisted of community-dwelling older Medicare beneficiaries with 68 years of age or older in 2018. ADRD diagnosis is based on the CMS Chronic Condition Warehouse (CCW) algorithm. CCW uses International Classification of Diseases, Ninth/Tenth Revision, Clinical Modification (ICD-9-CM/ICD-10-CM) codes from inpatient and outpatient claims in the three prior years, as described by the Centers for Medicare and

Medicaid Services Chronic Conditions Data Warehouse, accessed March 9, 2025 [21]. Under this algorithm, a beneficiary is classified as having ADRD if at least one qualifying diagnosis appears on a claim from any clinical setting included in CCW, including inpatient, skilled nursing facility, home health, Part B institutional, or carrier file. This method correctly identifies ~87% of patients with dementia using ICD-9-CM codes and has good discrimination with an area under the curve of 0.86 using ICD-10-CM codes. We identified MCI diagnosis using ICD-9-CM code 311.83 and ICD-10-CM code G31.84 [22–24]. ACP was identified using HCPCS billing code 99497 and 99498, which captures billable and clinician-documented ACP discussions.

We included beneficiaries with continuous Medicare A and B and no HMO enrolment during 2015–2018. Those with prior AWP or ACP in 2015–2017 were excluded. For beneficiaries who received an AWP, the index date was the calendar date of their first AWP in 2018. For the non-AWP cohort, an index month was randomly assigned to match the AWP cohort distribution, and Day 15 was set as the index date. Only beneficiaries with complete data were retained. (Table S1).

2.3 | Patient Characteristics

The beneficiary characteristics obtained for the analysis included demographics (age, gender identity, race/ethnicity, and residential area), original Medicare entitlement, Medicaid, and tobacco use that occurred in the 2 years before the index date. We categorized moderate/severe ADRD using a frailty index score ≥ 0.28 , consistent with the cut point identified in Park et al.'s validation study, which showed that this claims based frailty index can discriminate moderate to severe dementia (FAST stages 5–7) with a C statistic of 0.78 [25]. We extracted two county-level covariates (with quartile 1 the lowest and 4 the highest), which are physical inactivity and social activity; and one Zip code-level covariate: high school graduation rate [26–28]. County-level contextual variables were obtained from the CHR dataset and are widely used to capture environmental and socioeconomic conditions that may influence individual health behaviors and care utilization [29, 30]. Although county-level measures may introduce nondifferential misclassification, prior large-scale studies have shown that such contextual indicators remain strongly associated with individual health behaviors and preventive care use [31]. We included four medical service utilization characteristics in the 12 months before CI diagnosis: (1) number of hospitalizations (0, 1, 2, 3, ≥ 4); (2) number of neurologist visits; (3) number of psychiatrist visits; and (4) having a primary care provider (PCP). We divided individuals with ADRD/MCI into incident and prevalent cases. Patients with incident ADRD/MCI were first diagnosed in 2018, and patients with prevalent ADRD/MCI were first diagnosed before 2018.

2.4 | Outcome

The primary outcome was ACP initiation between the index date and December 31, 2021. Beneficiaries were followed from the AWP index date until ACP and censored at death, loss of Medicare Parts A/B coverage, enrollment in an HMO plan, or the end of the study period (12/31/2021), whichever occurred first. We treated death as a competing risk in all analyses.

2.5 | Statistical Analysis

To control for differences between AWP and non-AWP beneficiaries, we used propensity score matching. Propensity scores were estimated via logistic regression including comorbidities, age, gender, race/ethnicity, entitlement, dual eligibility, residence, education, physical inactivity, social activity, PCP, neurologist, psychiatrist visits, hospitalizations, and tobacco use. AWP month was exactly matched.

For each beneficiary with an AWP, we conducted 1:1 greedy nearest-neighbor propensity-score matching within a caliper of 0.2 standard deviations of the logit of the propensity score. Covariate balance before and after matching was assessed using standardized differences, with values below 0.10 indicating adequate balance. Patient characteristics are shown in Table 1.

The first analysis treated AWP exposure as a fixed baseline variable, using beneficiaries' AWP status in 2018 regardless of whether they subsequently continued or initiated AWP during follow-up. Beneficiaries were censored at loss of Medicare Parts A/B coverage, enrollment in an HMO plan, the end of follow-up (12/31/2021), or death, which was treated as a competing risk. A conditional Cox model estimated the association between AWP receipt in 2018 and ACP initiation.

In the second analysis, non-AWP beneficiaries were censored at their first AWP, and AWP beneficiaries were censored if no subsequent AWP occurred within 15 months. Additional censoring rules matched the first analysis. A conditional Cox model again estimated hazard ratios with 95% CIs.

In the third analysis, AWP was modeled as a time-dependent exposure, allowing transitions between AWP and non-AWP status. Participants were followed until ACP, insurance loss, or December 31, 2021, with death treated as a competing risk. An unconditional Cox proportional hazards model adjusted for all covariates in Table 1 was used to estimate the association between AWP status and ACP, with additional analyses stratified by baseline cognitive status. In this approach, AWP status was updated during follow up, and the time scale did not restart with each additional AWP.

All statistical analysis was performed using SAS/STAT software (SAS Institute Inc. SAS/STAT 9.4, SAS Institute Inc., Cary, NC, USA).

3 | Results

3.1 | Demographic Characteristics

Among 959,405 Medicare beneficiaries aged 68 years or older newly diagnosed with MCI or ADRD in 2018, 118,174 (12.3%) received an AWP. Before propensity-score matching, AWP recipients were more likely to be female, non-Hispanic White, and reside in metropolitan areas. Imbalances ($SD > 0.10$) were observed for metropolitan residence ($SD = 0.3$), having a primary care provider ($SD = 0.3$), neurologist visits ($SD = 0.1$), and dual eligibility ($SD = 0.1$), all more common among AWP recipients. AWP recipients were also more likely to have an identified primary care provider in the year before

TABLE 1 | Patient characteristics before propensity-score matching.

Patient characteristics	NO AWW	AWV	SD
<i>Previous MCI/ADRD</i>			
No	288,815 (34.3%)	44,026 (37.3%)	0.06
Yes	552,416 (65.7%)	74,148 (62.7%)	−0.06
<i>Cognitive status</i>			
MCI	88,924 (10.6%)	15,877 (13.4%)	0.09
Severe ADRD	112,263 (13.3%)	14,305 (12.1%)	−0.04
Mild ADRD	640,044 (76.1%)	87,992 (74.5%)	−0.04
<i>Sex</i>			
Female	515,878 (61.3%)	73,332 (62.1%)	−0.02
Male	325,353 (38.7%)	44,842 (37.9%)	0.02
<i>Age at diagnosis</i>			
68–74 years	183,812 (21.9%)	27,770 (23.5%)	0.04
75–79 years	171,561 (20.4%)	26,022 (22.0%)	0.04
80–84 years	186,227 (22.1%)	26,837 (22.7%)	0.01
≥85 years	299,631 (35.6%)	37,545 (31.8%)	−0.08
<i>Race</i>			
Non-Hispanic white	706,998 (84.0%)	99,932 (84.6%)	0.01
Hispanic	38,166 (4.5%)	5287 (4.5%)	0.00
Black	63,053 (7.5%)	8541 (7.2%)	0.01
Other	33,014 (3.9%)	4414 (3.7%)	0.00
<i>Location</i>			
Metro	622,725 (74.0%)	99,693 (84.4%)	0.26
Nonmetro	218,506 (26.0%)	18,481 (15.6%)	−0.26
<i>Education</i>			
<86.2	210,709 (25.0%)	25,550 (21.6%)	−0.08
86.2–91.0	208,542 (24.8%)	28,013 (23.7%)	−0.03
91.1–94.6	210,491 (25.0%)	30,395 (25.7%)	0.02
>94.6	211,489 (25.1%)	34,216 (29.0%)	0.09
<i>Original entitlement</i>			
OASI	747,053 (88.8%)	107,062 (90.6%)	0.06
DIB	92,738 (11.0%)	10,947 (9.3%)	−0.06
ESRD	735 (0.1%)	94 (0.1%)	0.00
DIB and ESRD	705 (0.1%)	71 (0.1%)	−0.01
<i>Dual eligibility</i>			
No	675,978 (80.4%)	101,092 (85.5%)	0.14
Yes	165,253 (19.6%)	17,082 (14.5%)	−0.14
<i>Physical_inactivity</i>			
<20.7	207,244 (24.6%)	30,593 (25.9%)	0.03

(Continues)

TABLE 1 | (Continued)

Patient characteristics	NO AWV	AWV	SD
20.7–24.3	206,470 (24.5%)	30,421 (25.7%)	0.03
24.4–28.1	212,602 (25.3%)	31,069 (26.3%)	0.02
> 28.1	214,915 (25.5%)	26,091 (22.1%)	–0.08
<i>Social activity</i>			
< 7.2	190,783 (22.7%)	30,149 (25.5%)	0.07
7.2–9.2	214,477 (25.5%)	32,470 (27.5%)	0.04
9.3–11.6	196,434 (23.4%)	28,661 (24.3%)	0.02
> 11.6	239,537 (28.5%)	26,894 (22.8%)	–0.13
<i>Having PCP</i>			
No	430,471 (51.2%)	42,683 (36.1%)	–0.31
Yes	410,760 (48.8%)	75,491 (63.9%)	0.31
<i>Neurologists visit</i>			
No	668,525 (79.5%)	88,451 (74.8%)	–0.11
Yes	172,706 (20.5%)	29,723 (25.2%)	0.11
<i>Psychiatrics visit</i>			
No	810,469 (96.3%)	112,689 (95.4%)	–0.05
Yes	30,762 (3.7%)	5485 (4.6%)	0.05
<i>Number of hospitalizations</i>			
0	534,693 (63.6%)	78,772 (66.7%)	0.01
1	185,250 (22.0%)	24,831 (21.0%)	–0.02
2	70,303 (8.4%)	8695 (7.4%)	–0.04
3	28,322 (3.4%)	3395 (2.9%)	–0.04
4+	22,663 (2.7%)	2481 (2.1%)	–0.04
<i>Tobacco</i>			
No	770,820 (91.6%)	108,440 (91.8%)	0.00
Yes	70,411 (8.4%)	9734 (8.2%)	0.00

diagnosis (63.9% vs. 48.8%) and to live in counties with higher high school graduation rates (Table 1).

After 1:1 propensity score matching based on a comprehensive set of sociodemographic, clinical, and health care utilization characteristics, all baseline covariates were well balanced between the two groups, with standardized differences below 0.10 for every variable (Table S2).

3.2 | Main Analysis

In the propensity-score-matched cohort followed from the 2018 index date through December 31, 2021, receipt of an AWV was significantly associated with higher ACP initiation. Cumulative incidence curves diverged within 6 months and remained separated throughout follow-up (Figure 1). In a conditional Cox model accounting for death as a competing risk, AWV receipt

was associated with a higher hazard of ACP initiation compared with no AWV (HR, 1.86 [95% CI, 1.84–1.89]) (Figure 2); no additional covariate adjustment was performed because baseline characteristics were balanced after matching.

ACP uptake among AWV recipients occurred predominantly on the day of the AWV (9.00%), with modest increases to 9.41% within 30 days, 14.78% within 1 year, and 27.62% within 4 years. In contrast, ACP initiation was rare in the non-AWV group on the index date (0.01%) and reached 17.46% by 4 years (Table S3).

3.3 | Sensitivity Analyses

During the follow-up period, in the non-AWV group ($n = 841,231$), 27,380 received an AWV during the follow-up period, and 2547 of them completed ACP on the same day. In the AWV group ($n = 118,174$), 19,098 received an AWV only in 2019,

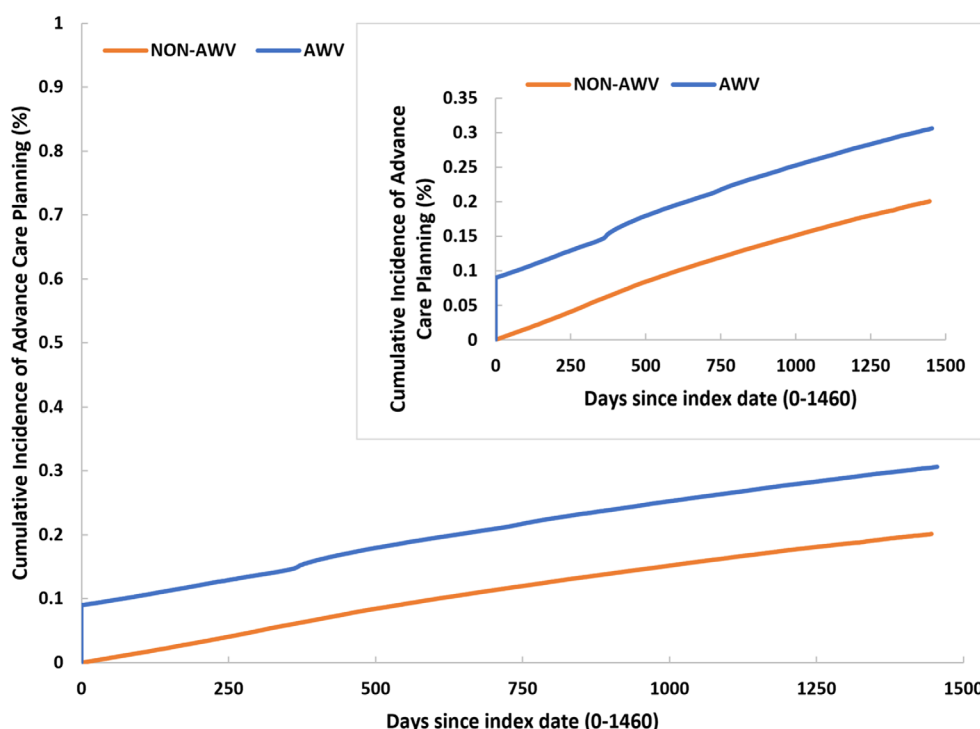


FIGURE 1 | Cumulative incidence of advance care planning (ACP) completion among beneficiaries with and without an annual wellness visit (AWV) at baseline. This figure shows the cumulative incidence of ACP completion over 1460 days following the index date, comparing beneficiaries who received an AWV with those who did not. The main panel displays cumulative incidence estimates using the full y-axis scale starting from zero, reflecting the small absolute differences between groups. An embedded inset plot is included to facilitate visualization of differences at low incidence levels without altering the overall scale. The x-axis represents days since the index date, and the y-axis represents cumulative incidence (%). ACP, advance care planning; AWV, annual wellness visit. Values shown represent cumulative incidence estimates.

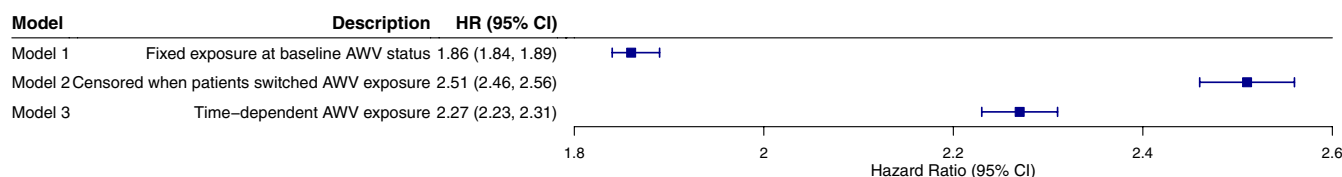


FIGURE 2 | Hazard ratios for the association between annual wellness visit (AWV) receipt and advance care planning (ACP) initiation in the primary analysis and two sensitivity analyses. This figure displays hazard ratios and 95% confidence intervals for three analytic approaches assessing the association between AWV receipt and initiation of Advance care planning (ACP). Model 1 represents the primary analysis using AWV status at baseline. Model 2 applies censoring rules to account for AWV initiation among beneficiaries initially classified as non-AWV and for discontinuation among AWV recipients. Model 3 treats AWV receipt as a time-dependent exposure. Hazard ratios greater than 1 indicate a higher likelihood of ACP initiation among AWV recipients compared with non-recipients. ACP, advance care planning; AWV, annual wellness visit; CI, confidence interval; HR, hazard ratio.

9480 had AWVs in both 2019 and 2020 (but not in 2021), and 9947 received AWVs in all 3 years, from 2019 to 2021.

In clinical practice, AWV participation can change over time as some beneficiaries begin AWVs during follow-up while others discontinue them. To reflect this variability in exposure, we conducted two sensitivity analyses. The first applied censoring rules when AWV status changed, and the second treated AWV receipt as a time-dependent covariate to capture evolving patterns of AWV use.

In the first sensitivity analysis, censoring non-AWV beneficiaries at first AWV receipt and AWV recipients without a subsequent AWV within 15 months strengthened the association

between AWV and ACP initiation (HR, 2.51 [95% CI, 2.46–2.56]) (Figure 2).

In the second sensitivity analysis, a Cox model treating AWV as a time-dependent covariate and adjusting for all baseline covariates produced consistent results (HR, 2.27 [95% CI, 2.27–2.31]), with death treated as a competing risk (Figure 2). Higher ACP use was also independently associated with AD RD severity and age, including severe AD RD (HR, 1.28 [95% CI, 1.23–1.32]), mild AD RD (HR, 1.09 [95% CI, 1.06–1.12]), and age ≥ 85 years versus 68–74 years (HR, 1.30 [95% CI, 1.27–1.33]).

Differences were also observed by race, geography, and socioeconomic characteristics. Black beneficiaries had higher ACP

use than non-Hispanic White beneficiaries (HR, 1.16 [95% CI, 1.13–1.20]), whereas Hispanic beneficiaries had lower ACP use (HR, 0.91 [95% CI, 0.88–0.96]). Living in a nonmetropolitan area was associated with significantly lower ACP initiation (HR, 0.82 [95% CI, 0.79–0.84]). Dual eligibility (HR, 1.13 [95% CI, 1.10–1.16]), having a primary care provider (HR, 1.03 [95% CI, 1.01–1.05]), and neurologist consultation (HR, 1.04 [95% CI, 1.02–1.06]) were each associated with increased ACP use. A dose–response pattern was observed for hospitalization: beneficiaries with four or more hospitalizations had the greatest likelihood of ACP completion (HR, 1.48 [95% CI, 1.40–1.56]) compared with those with no hospitalizations (Table S4).

Significant interactions were observed between AWV and race, geography, sex, and education. Stratified analyses showed consistent associations between AWV and ACP use across subgroups, with the strongest associations among Hispanic beneficiaries (HR, 2.78 [95% CI, 2.55–3.03]) and residents of nonmetropolitan areas (HR, 3.34 [95% CI, 3.16–3.52]). Associations were similar by sex (female: HR, 2.34 [95% CI, 2.29–2.39]; male: HR, 2.15 [95% CI, 2.09–2.22]). Stage-stratified analyses under Method 3 also showed consistent associations across cognitive stages (MCI: HR, 2.16; mild ADRD: HR, 2.37; severe ADRD: HR, 1.91), with attenuated magnitude at higher education levels (Table S5).

4 | Discussion

This cohort study assessed the association between receipt of an AWV in 2018 and the timing of first ACP billing over a 4-year follow-up period among older Medicare beneficiaries newly diagnosed with MCI or ADRD. We found that 27.62% of AWV recipients initiated ACP within 4 years compared with 17.46% of those without an AWV (HR, 1.86 [95% CI, 1.84–1.89]). In other words, patients who received an AWV were more likely to complete ACP than those who did not. Sensitivity analyses yielded consistent results, including a stronger association when AWV was treated as a time-varying exposure and when the cohort was restricted to patients with continued AWV use. Stage-stratified results echoed this pattern. The association between AWV and ACP was present at all stages, with relatively lower magnitude in severe ADRD. These findings highlight the potential value of AWVs as an opportunity to promote timely ACP engagement among older adults across different dementia stages.

Independent of AWV receipt, ACP completion was more common in both mild and severe ADRD than in MCI, with the highest uptake in severe disease. This pattern likely reflects ACP being initiated during periods of significant functional decline or acute medical crises, consistent with frontline clinical practice and decisions made under time pressure [32, 33]. At the severe stage, decision-making capacity is often reduced, and discussions depend more on surrogates, which may limit concordance with patient values without prior early conversations [34, 35]. These findings support initiating ACP earlier in the dementia trajectory, when patients can still articulate preferences and plans can be iteratively updated with families and clinicians [33].

AWVs have been widely shown to increase the use of preventive services among older adults, and the introduction of ACP

billing codes into AWV reimbursement likely enhanced this effect [19, 36]. Because the AWV requires assessment of cognition, function, and future health risks, clinicians often address future medical decision making during the visit, which can naturally lead to ACP discussions. In our study, 27.6% of AWV recipients completed an ACP discussion, a rate substantially higher than the 21.9% reported in the general population [14]. This elevated rate may reflect our focus on beneficiaries with MCI or dementia, a population for whom ACP is especially relevant. Prior research suggests that ACP, as part of a broader palliative care approach, can be particularly beneficial for individuals with cognitive decline [12]. However, many of these patients lack access to palliative care services and live outside of institutional care settings, limiting opportunities for structured ACP conversations [37]. The 2016 Medicare reimbursement change, which allowed ACP to be billed as an optional component of the AWV, may have addressed this gap by creating a mechanism for timely planning in outpatient settings [38, 39]. To our knowledge, this is the first study to quantify this association in a MCI/ADRD patient population, and our observed hazard ratio of 1.86 indicates that at any point during follow-up, AWV recipients were 86% more likely to initiate ACP compared to non-recipients, highlighting the substantial impact of AWV receipt on ACP uptake.

Although same-day ACP was the most common timing among AWV recipients who ultimately completed ACP, only 9.0% of all AWV recipients did so on the day of the wellness visit, indicating substantial room for improvement in leveraging the AWV as a same-day ACP opportunity. While Medicare policy permits ACP to be billed as part of the AWV, same-day completion may be hindered by logistical and operational constraints [40, 41]. Existing billing regulations typically require physician involvement, which can be difficult to coordinate within the time-limited structure of a wellness visit [40, 42]. Clinicians have reported barriers such as inadequate training, low visibility of the ACP code, and complex documentation requirements, all of which may discourage initiating AWV in the same day with ACP [38, 40, 43]. When multiple preventive care tasks must be addressed in a single visit, providers may deprioritize ACP discussions due to their time intensity. Structural strategies described in the literature include expanding billing authority to nonphysician staff such as clinical social workers and applying implementation frameworks like PRISM to redesign workflows and support task delegation [38, 44, 45].

Our subgroup analysis revealed that the positive association between AWV and ACP uptake was greater among underserved populations. Interactions with race or ethnicity, geography, sex, and education were statistically significant, with larger gains among residents of nonmetropolitan areas and among Hispanic beneficiaries. While nonmetropolitan and Hispanic beneficiaries were generally less likely to engage in ACP, consistent with previous studies that documented disparities due to limited access, lower health literacy, and cultural or linguistic barriers, those who received an AWV experienced greater improvements in ACP uptake compared to their respective counterparts [46, 47]. The AWV may help mitigate these disparities by providing a structured and reimbursable opportunity for ACP discussions, particularly for individuals who have historically lacked access to comprehensive primary care [16]. From a health equity perspective, expanding access to AWVs may serve as a feasible

pathway to reduce disparities in end-of-life planning for cognitively impaired populations.

5 | Clinical Implication

Our findings suggest that the AWV can serve as a system-level reminder, especially if incorporated into the health-system electronic health record to initiate ACP discussions among cognitively impaired older adults. Since AWV is typically scheduled while patients still retain decision-making capacity, it provides a timely and structured opportunity to introduce ACP conversations. This system-level approach could be complemented by involving interdisciplinary team members, such as nurses or social workers, to support and continue ACP engagement beyond the initial conversation [44]. To increase uptake, clinics may include brief, plain-language ACP materials with AWV invitations or portal reminders and add a simple yes/no item to the AWV questionnaire or health risk assessment to assess interest in ACP. These strategies can prime patients and caregivers before the visit and facilitate same-day ACP when appropriate.

6 | Strengths and Limitations

This study has several strengths. It leveraged a nationally representative sample of Medicare beneficiaries, focusing specifically on individuals with MCI/ADRD. The use of multiple analytic models, including propensity score matching and sensitivity analyses, enhanced the robustness of our findings and helped mitigate potential confounding.

Several limitations should be acknowledged. First, ACP was identified using HCPCS codes 99497–99498 (Table S1), which capture only billed and clinician-documented encounters and likely underestimate ACP occurrence. ACP discussions may not be captured in claims due to bundled payments, documentation practices, clinician unfamiliarity with billing, or involvement of non-physician team members. Moreover, claims data do not capture the content, timing, or quality of ACP discussions, reflecting billing activity rather than the depth of ACP conversations. Our findings apply only to older adults enrolled in traditional Fee-For-Service Medicare; ACP uptake may differ among beneficiaries enrolled in Medicare HMO plans. At the same time, the identification of MCI/ADRD was based on diagnosis codes from administrative claims, which may be subject to sensitivity or misclassification issues [41, 48].

7 | Future Work

Although prior studies using electronic health record data have examined ACP in older adults, few focused on the AWV context or patients with MCI/ADRD. Future research should leverage EHR data to validate whether ACP billed during AWVs reflects actual discussions and to assess their content and quality. Additional work is needed to examine associations with end-of-life care utilization, the role of community support in nonmetropolitan areas, and concordance with patient preferences. System-level interventions are also needed to improve AWV workflows and support timely ACP, including enhanced same-day completion

and integration of decision support tools or clinician training informed by clinician, patient, and caregiver input.

Author Contributions

Y.-F.K., M.A.R., H.-M.T., and S.O. conceived the project, designed the methods, and created the quantitative analysis plan. Y.S. conducted the analysis of the data. Z.H. and Y.S. wrote the manuscript. Y.-F.K., Z.H., M.A.R., Y.S., H.-M.T., and S.O. revised the manuscript and contributed significantly to its intellectual content, interpretation of the results, and discussion. All authors approved of its final version.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Cohort selection process for identifying eligible Medicare beneficiaries with MCI or ADRD. **Table S2:** Full characteristics of AWV and non-AWV recipients before and after propensity score matching. **Table S3:** Cumulative incidence of advance care planning (ACP) completion at selected follow-up intervals by AWV status. **Table S4:** Full Cox proportional hazards model for Method 3 including all covariates. **Table S5:** Stratified analyses evaluating the association between AWV receipt and ACP initiation across demographic and clinical subgroups.